Amazon Review Intelligence Dashboard & Sentiment Prediction

Complete Project Documentation

Built with Claude AI (claude-sonnet-4-6) · Python · HTML/JavaScript · Chart.js

GitHub: github.com/oping000/Claude-Projects

Component	Description
HTML Dashboard	Interactive browser app — analyze reviews live with AI
Python Pipeline	Batch process full CSV files through Claude API
Sentiment Analysis	Classifies reviews as Positive, Neutral, or Negative
Predictions Engine	Escalation risk, star rating prediction, churn signal
Product Health	Tracks sentiment trends per product over time
CSV Export	Download enriched results for Power BI or Excel

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1. Project Overview

The Amazon Review Intelligence Dashboard is an end-to-end AI-powered pipeline that transforms unstructured Amazon product reviews into structured, actionable business intelligence. It was built using Claude AI (claude-sonnet-4-6), Python, and vanilla HTML/JavaScript.

The core business problem it solves: companies receive thousands of unstructured customer reviews daily and need to instantly understand customer sentiment, predict escalation risk, and identify product health trends — without reading every review manually.

What the pipeline extracts from every review:

Output Field	What It Tells You
Sentiment	Is the customer Positive, Neutral, or Negative?
Sentiment Score	How positive/negative on a scale of 1-10?
Issue Category	Quality, Value, Performance, Shipping, etc.
Priority	How urgently does this need attention?
Summary	One clean sentence summarizing the review
Predicted Stars	What star rating would this customer give?
Rating Confidence	How confident is the model in that prediction?
Escalation Risk	Will this customer demand a refund or go public?
Escalation Probability	Percentage likelihood of escalation (0-100%)
Churn Signal	Will this customer ever buy again?
Churn Probability	Percentage likelihood of churn (0-100%)

2. Architecture & How It Works

The pipeline follows a straightforward extract-process-structure pattern:

Step 1 — Input

Unstructured review text is passed in — either pasted manually into the HTML dashboard or read from a CSV file by the Python script.

Step 2 — Claude API Call

The review text is sent to the Claude AI API (claude-sonnet-4-6) with a carefully engineered prompt that forces a guaranteed JSON schema response.

Step 3 — Structured Output

Claude returns clean JSON with all fields: sentiment, score, category, priority, predictions, and reasoning.

Step 4 — Display or Export

The HTML dashboard displays results visually with charts and badges. The Python script writes results to an enriched CSV file.

Why Claude AI instead of traditional NLP?

Traditional sentiment tools use keyword matching or simple ML models. Claude understands nuance, context, and intent — so a review saying "not bad" is correctly identified as mildly positive, and "works fine I guess" is correctly flagged as lukewarm rather than enthusiastic.

3. Project Files

File	Type	Purpose
amazon_review_intelligence_dashboard.html	HTML	Full interactive dashboard — all 5 tabs
amazon_sentiment_pipeline.py	Python	Batch processing pipeline — CSV in, enriched CSV out
amazon_sentiment_analysis.csv	CSV	Sample input — 30 Amazon electronics reviews
amazon_enriched_output.csv	CSV	Pipeline output — all reviews enriched with AI analysis
product_health_report.csv	CSV	Per-product health summary generated by Python script
.env	Config	Your API key — never uploaded to GitHub
.env.example	Config	Template showing the .env format — safe to share
.gitignore	Config	Blocks sensitive files from being uploaded to GitHub
README.md	Docs	Project description shown on GitHub

4. Sentiment Analysis — Fields Explained

Sentiment Label

Claude classifies each review into one of three categories based on the overall emotional tone of the text:

Label	Meaning	Typical Score Range
Positive	Customer is satisfied, enthusiastic, or recommending	7 - 10
Neutral	Mixed feelings, neither clearly happy nor unhappy	4 - 6
Negative	Customer is dissatisfied, frustrated, or complaining	1 - 3

Sentiment Score (1-10)

A numeric score representing how positive or negative the review is. Unlike the label which has only 3 values, the score gives finer-grained insight:

Score Range	Interpretation
8 - 10	Highly positive — enthusiastic customer, likely to recommend
6 - 7	Mildly positive — satisfied but not glowing
4 - 5	Neutral — mixed signals, some positives and negatives
2 - 3	Mildly negative — disappointed but not angry
1	Highly negative — very unhappy, likely to escalate

Issue Category

Claude tags each review with the primary topic the customer is discussing:

Category	What it covers
Quality	Build quality, materials, durability, defects
Value	Price, worth the money, compared to alternatives
Performance	Speed, charging rate, data transfer, output
Durability	Long-term reliability, wear and tear
Shipping	Delivery speed, packaging, arrival condition
Customer Service	Support responsiveness, refunds, replacements
Compatibility	Works with devices, connectors, operating systems

General	Overall impression without a specific focus

Priority

How urgently the review needs business attention:

5. Predictions Engine — Fields Explained

The predictions engine goes beyond analyzing what the customer said — it predicts what they are likely to do next. This is the core business value of the pipeline.

Predicted Star Rating

Claude reads the review text and predicts what star rating the customer would give, independent of any actual rating provided. This is useful for:

- Detecting fake or manipulated reviews — if a customer writes a very negative review but gives 5 stars, the predicted rating will flag the mismatch
- Filling in missing ratings — some review platforms allow text without a star rating
- Validating existing ratings — confirms whether the written sentiment matches the numeric rating

Rating Confidence (0-100%)

How confident Claude is in the predicted star rating. Higher confidence means the review language was clear and consistent:

Escalation Risk & Probability

Predicts whether the customer is likely to escalate — meaning they will demand a refund, leave a public complaint, contact support aggressively, or post on social media about the product.

Important note: within each risk level there is still a spectrum. A "Low" review with 22% probability is more at-risk than a "Low" review with 5% probability. Both are unlikely to escalate, but the 22% one is worth watching.

Churn Signal & Probability

Predicts whether the customer will ever purchase from this seller or brand again.

The most actionable insight: a "Moderate" churn signal with 45% probability is the customer you should send a discount code to before they leave. They are right on the edge and a small gesture could retain them.

6. Product Health Detection

The Product Health tab analyzes multiple reviews for the same product and detects whether overall sentiment is stable, declining, or at risk. This is prediction #2 from the original project roadmap.

How to use it

- Enter the product name in the product name field
- Paste one separate customer review per line in the text area — each line must be a different customer's review
- Order the reviews from oldest to newest so Claude can detect trend direction
- Click Analyze Product Health

What it returns

Status definitions

7. Understanding the CSV Output

When you run the Python pipeline, it produces amazon_enriched_output.csv with these columns:

Column	Type	Description
product_id	String	Original Amazon product identifier
product_name	String	Product name (truncated to 80 chars)
product_category	String	Category from the original data
star_rating	Float	Original star rating from Amazon (1-5)
sentiment	String	Positive, Neutral, or Negative
sentiment_score	Integer	Score from 1-10 (10 = most positive)
issue_category	String	Quality, Value, Performance, etc.
priority	String	High, Medium, or Low
key_issue	String	Main topic in 2-4 words
summary	String	One-sentence summary under 20 words
predicted_stars	Float	Claude's predicted star rating (1.0-5.0)
rating_confidence	Integer	Confidence in predicted stars (0-100)
rating_reasoning	String	Explanation of the star prediction
escalation_risk	String	High, Medium, or Low
escalation_probability	Integer	Probability of escalation (0-100)
escalation_reasoning	String	Explanation of escalation prediction
churn_signal	String	Strong, Moderate, or Weak
churn_probability	Integer	Probability of churn (0-100)
churn_reasoning	String	Explanation of churn prediction
review_title	String	Cleaned review title (120 chars max)
review_content	String	Cleaned review content (250 chars max)

8. How Sentiment Score & Star Rating Correlate

A common question when reviewing the CSV output is how the sentiment score relates to the original star rating. Here is the relationship:

General pattern

- Higher star rating → higher sentiment score → Positive sentiment label
- Lower star rating → lower sentiment score → Neutral or Negative sentiment label

Why the same star rating can produce different sentiment scores

Claude does not simply convert the star rating into a score. It reads the actual review text. This means two products both rated 3.9 stars can receive different sentiment scores if their review language differs. One might use words like "works fine, no complaints" (scoring 6) while another says "disappointed, feels cheap" (scoring 4).

This is exactly what makes the pipeline more valuable than sorting by star rating alone — it captures the nuance and intent behind the words.

Star Rating	Typical Sentiment Score	Typical Sentiment Label
4.5 - 5.0	8 - 10	Positive
3.8 - 4.4	6 - 8	Positive or Neutral
3.0 - 3.7	4 - 6	Neutral
2.0 - 2.9	2 - 4	Negative
1.0 - 1.9	1 - 2	Negative

9. How Escalation Risk & Probability Work Together

A question that came up during this project: if all the escalation risks show Low, why do the probabilities range from 5% to 22%? The answer is that they are both correct — they are just measuring different things.

The label is a category. The probability is a precise number within that category.

Think of it like a traffic light. "Green" means go — but some green lights are at a busy intersection and some are on an empty street. The label is the color. The probability is the actual traffic volume.

How to use both fields together

- Low + 5% — Happy customer, no action needed
- Low + 22% — Still low risk but worth a quick check
- Medium + 45% — Proactive outreach recommended
- High + 80% — Urgent response required

A truly mismatched result would be Low + 75% or High + 10% — those would indicate a model error. Low + 22% is intentional and correct.

10. How Churn Signal & Probability Work Together

The churn signal and probability follow the same logic as escalation — the signal is the category label, the probability is the precise percentage within that category.

The most actionable customer in your data

A Moderate churn signal with 45% probability is the highest-priority retention target. This customer has not fully decided to leave — they are right on the edge. A small gesture (discount, apology, replacement offer) at this point is far cheaper than acquiring a new customer.

Combining churn with escalation

The most dangerous customer profile in your data is High escalation risk + Strong churn signal. This customer is not only likely to complain publicly but also unlikely to ever return. These reviews should be flagged for immediate response by the customer service team.

11. Setting Up & Running the Python Pipeline

Requirements

```
pip install anthropic pandas python-dotenv
```

Folder structure

```
sentiment-pipeline/ ■■■ amazon_sentiment_pipeline.py ■■■
amazon_sentiment_analysis.csv ■■■ .env ← your real API key (never on GitHub) ■■■
.env.example ← safe template ■■■ .gitignore ← blocks .env from GitHub
```

Creating your .env file

Never paste your API key directly into the Python file. Instead create a .env file in the same folder with this content:

```
ANTHROPIC_API_KEY=sk-ant-your-actual-key-here
```

On Windows PowerShell, create the file with the correct UTF-8 encoding using:

```
python -c "open('.env', 'w',
encoding='utf-8').write('ANTHROPIC_API_KEY=sk-ant-yourkey\n')"
```

Running the pipeline

```
python amazon_sentiment_pipeline.py
```

The script will process reviews one by one, showing progress in the terminal. When complete it asks if you want to run the product health check per product.

Key settings in the script

Setting	Default	Description
INPUT_CSV	amazon_sentiment_analysis.csv	Your input file name
OUTPUT_CSV	amazon_enriched_output.csv	Where results are saved
MODEL	claude-sonnet-4-6	Claude model to use
MAX_REVIEWS	30	Set to None to process all rows
DELAY_SEC	0.5	Pause between API calls to avoid rate limits

12. Using the HTML Dashboard

The HTML dashboard is a standalone file that runs entirely in your browser. No server, no install, no dependencies beyond a modern browser and an Anthropic API key.

How to open it

- Download amazon_review_intelligence_dashboard.html to your computer
- Double-click the file — it opens in your default browser
- Paste your Anthropic API key in the API Key box at the top
- Start analyzing reviews

The five tabs

Important notes

- The dashboard is session-based — closing the browser tab clears all history. Use Download CSV before closing.
- Your API key is never stored — it lives only in memory for the current session.
- The Product Health tab requires at least 2 separate reviews on separate lines — not one review split across lines.

13. API Key Security

Protecting your Anthropic API key is critical. A leaked key can be used by others to run API calls billed to your account.

In the HTML dashboard

- The key is typed into a password field and lives only in browser memory
- It is never written to disk, localStorage, or any external service
- Closing the browser tab clears it completely

In the Python pipeline

- The key lives in the .env file on your local machine
- The .gitignore file blocks .env from ever being uploaded to GitHub
- The .env.example file shows the format safely without containing a real key
- Never paste your key directly into the Python script file

If your key is ever exposed

- Go immediately to console.anthropic.com
- Click API Keys in the sidebar
- Delete the compromised key and create a new one

14. GitHub Setup & Deployment

The project is hosted at: github.com/oping000/Claude-Projects under the amazon-sentiment-pipeline subfolder.

Pushing updates via GitHub Desktop (recommended)

- Open GitHub Desktop
- Select the Claude-Projects repository
- Copy updated files into the amazon-sentiment-pipeline folder on your computer
- GitHub Desktop will automatically detect the changed files
- Type a commit message describing what changed
- Click Push origin

What is and is not on GitHub

File	On GitHub?	Reason
amazon_review_intelligence_dashboard.html	Yes	Safe — no keys in the file
amazon_sentiment_pipeline.py	Yes	Safe — reads key from .env
amazon_sentiment_analysis.csv	Yes	Sample data, no sensitive info
README.md	Yes	Project documentation
.env.example	Yes	Safe template with placeholder
.gitignore	Yes	Required for protection
.env	NO	Contains your real API key
amazon_enriched_output.csv	NO	Blocked by .gitignore

15. Future Enhancements

The following features were identified during the project build as natural next steps:

Predictions not yet built

The original project roadmap included 5 predictions. Three are fully built. Two remain:
